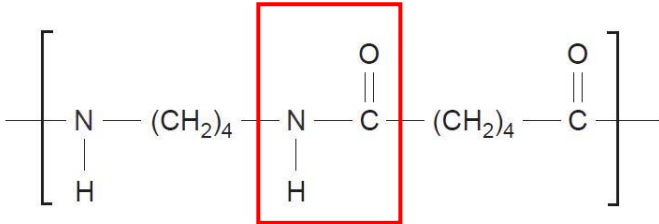


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		award (1) for correct product NH ₃ award (1) for correct balancing (only if product is correct) 2 NaOH 2 NH ₃		2		2		
		(ii)		$0.075 \times \frac{60}{100} = 0.045$ mol of the diamine formed (1) mass of diamine = $0.045 \times 88 = 3.96$ g (1)	1	1		2	1	
		(iii)			1			1		
		(iv)	I	award (1) for any of following only one stage involved therefore quicker only one stage involved therefore probably higher percentage yield fewer waste products therefore greener			1	1		
			II	award (1) for either of following no more effervescence / bubbles (of nitrogen) evolved no more precipitation of the acid			1	1		1